

Total No. of Questions : 12]

SEAT No. :

P3151

[Total No. of Pages : 2

[4858]-1008
T.E. (Civil) (Semester - II)
ENVIRONMENTAL ENGINEERING - I (End - Sem)
(2012 Pattern)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10, Q11 or Q12*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks*
- 4) *Assume suitable data, if necessary.*
- 5) *Use of scientific calculator is allowed*

Q1) Discuss the sources and effects of noise pollution. **[6]**

OR

Q2) Explain the principle and working of settling chamber for removing particulate matter. **[6]**

Q3) Explain with neat sketch the working, location and function of river and canal intake. **[6]**

OR

Q4) Explain the factors affecting the rate of demand. **[6]**

Q5) Draw the flow diagram/layout of a water supply scheme using Rapid sand filter .Write clearly the purpose of each unit in the water supply scheme. **[8]**

OR

Q6) Design a tube settler module with the following data- **[8]**

- a) Average output required from tube settler = $250\text{m}^3/\text{hr}$.
- b) Loss of water in desludging = 2% of output required.
- c) Average design flow = $(250 \times 100) / (100 - 2) = 255.1\text{m}^3/\text{hr}$.
- d) Cross section of square tubes — $50\text{mm} \times 50\text{mm}$.
- e) Length of tubes = 1m.
- f) Angle of inclination of tubes 60°

P.T.O.

Q7) Design a clariflocculator for desired average outflow of $250\text{m}^3/\text{hr}$, water lost in desludging - 2%, design average flow = $(250 \times 100)/(100-2) = 255.1 \text{ m}^3/\text{hr}$, detention period - 20 minutes and average value of velocity gradient $G = 40/\text{second}$. **[16]**

OR

Q8) a) Explain the theory of filtration. **[4]**
b) With a neat sketch explain back washing of rapid sand gravity filter. **[6]**
c) What is coagulation & flocculation? Draw a neat sketch of a flocculator. **[6]**

Q9) a) Explain break point chlorination. Define disinfection and list the different types of disinfectants used. **[8]**
b) Chlorine usage in treatment plant of 20 MLD of water is $8.5\text{kg}/\text{day}$. The residual chlorine content after 10min. is $0.2\text{mg}/\text{L}$ calculate on dosage of chlorine in mg/L and chlorine demand of water. **[4]**
c) State the factors affecting disinfection **[4]**

OR

Q10) a) Explain demineralization of water by Reverse Osmosis method. **[6]**
b) Discuss colour&odour removal by adsorption. **[6]**
c) Explain fluoridation & defluoridation of water. **[4]**

Q11) a) Differentiate between continuous & intermittent system of water supply. **[6]**
b) Explain detection and prevention of wastage of water. **[6]**
c) With a neat sketch explain dead end & reticulated distribution system. **[6]**

OR

Q12) a) Explain the benefits of rain water harvesting and discuss the different methods of rain water harvesting. **[9]**
b) Discuss the concept of packaged WTP for townships **[9]**

